LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 2 Test (GSEB pattern): Acids, Bases and Salts

Chapter / Topic: Acids, Bases and Salts

For teacher use only. Do not distribute to students.

SECTION A

Q1. A solution turns red litmus blue. Its pH is most likely to be: **[1]**

Answer: (d) 10. A solution that turns red litmus blue is basic, so its pH is more than 7.

Q2. Which of the following gives carbon dioxide with dilute hydrochloric acid? **[1]**

Answer: (a) Sodium hydrogencarbonate. $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$

Q3. Which gas is given off when dilute hydrochloric acid is added to sodium carbonate? **[1]**

Answer: (b) Carbon dioxide. $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

Q4. Which of these has the highest pH? **[1]**

Answer: (c) Milk of magnesia (it is basic)

Q5. The chemical formula of bleaching powder is: **[1]**

Answer: (c) $CaOCl_2$

Q6. Which indicator turns pink in a basic solution? **[1]**

Answer: (d) Phenolphthalein

Q7. Which of the following is used as an antacid? **[1]**

Answer: (b) Milk of magnesia, $Mg(OH)_2$, a mild base that neutralises excess stomach acid.

Q8. The chemical formula of Plaster of Paris is: **[1]**

Answer: (a) $CaSO_4 \cdot \frac{1}{2}H_2O$

Q9. Plaster of Paris is obtained by heating: **[1]**

Answer: (c) gypsum ($CaSO_4 \cdot 2H_2O$), heated at 373 K, which gives $CaSO_4 \cdot \frac{1}{2}H_2O$.

SECTION B

Q10. Why does dry HCl gas not change the colour of dry blue litmus paper? **[2]**

Answer: Acids show acidic behaviour only in the presence of water, because H^+ ions (H_3O^+) are produced only in solution. Dry HCl gas does not dissociate into H^+ ions, so it does not affect dry litmus.

- Acids need water to produce H^+ ions: 1
- Dry HCl does not form H^+ ions: 1

Q11. What is a universal indicator? Why is it more useful than litmus? **[2]**

Answer: A universal indicator is a mixture of several indicators that shows different colours at different pH values (different concentrations of H^+ ions). Litmus only tells whether a solution is acidic or basic. A universal indicator also tells how strong the acid or base is (its pH).

- Definition: 1
- Shows strength / pH: 1

Q12. Write the chemical name and formula of baking soda. Write the equation for the reaction that takes place when it is heated. **[2]**

Answer: Sodium hydrogencarbonate, $NaHCO_3$. On heating: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

- Name and formula: 1
- Balanced equation: 1

SECTION C

Q13. A student tests three solutions P, Q and R with universal indicator and finds their pH to be 2, 7 and 12 respectively. **[3]**

(i) Q (pH 7) [1]
(ii) P. The lower the pH, the higher the H^+ ion concentration. [1]
(iii) R (pH 12), because it is basic. [1]

Q14. A solution has a pH of 3. **[3]**

(i) Acidic [1]
(ii) The pH increases towards 7, because the H^+ ion concentration decreases. [1]
(iii) The pH rises until it reaches 7 (neutral). [1]

SECTION D

Q15. Tooth enamel, made of calcium hydroxyapatite (a form of calcium phosphate), is the hardest substance in the body. It does not dissolve in water, but… **[4]**

(i) 5.5 [1]
(ii) To neutralise the excess acid in the mouth and prevent tooth decay. [1]
(iii) Bacteria break down sugar and food particles left in the mouth and produce acids. Prevention (any two): brush after eating, rinse the mouth after meals, avoid sticky sweets, cut down on sugary snacks. [2]

- Source of acid: 1
- Two ways: ½ + ½